

Foundations of Artificial Intelligence in Health

Thursdays, 3:30–6:30pm · 3600 Civic Center Blvd, Room 6E 031

August 27 – December 3, 2026

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Welcome to BMIN 5200. This syllabus outlines the logistical details of the course. All course materials live on the [course website](#); Canvas is used only for submitting assignments, viewing grades, and watching class recordings.

Don't hesitate to ask if you have any questions about the information here or elsewhere — Joe can be reached at joseph.romano@pennmedicine.upenn.edu, or you can ask before or after class.

Course description

As a subfield of computer science, artificial intelligence (AI) is often used interchangeably with the term ‘machine learning’, which is itself only a subfield of AI dealing with the broader concept of *inductive reasoning*. However, a wealth of key prerequisite topics that focus on *deductive reasoning* cover a large portion of biomedical informatics applications being actively used today. These founding principles of AI and their intersection with biomedical informatics are the focus of this first course on AI, and lay the groundwork for future coursework on machine learning, artificial neural networks, and generative AI.

This course is divided into modules that cover (1) introductory/background materials, (2) knowledge representation, (3) logic, (4) essentials of rule-based systems, (5) search, (6) information structure and inference, and (7) special topics. These topics offer a global foundation of branches of AI application and research in biomedical domains, including concepts that will later support a deeper understanding of inductive reasoning and machine learning.

In a practical sense, this course focuses on how biomedical data can be organized, represented, interpreted, searched, and applied in order to derive knowledge, make decisions, and ultimately make predictions while mitigating bias.

To reiterate: this course does not focus heavily on machine learning or generative AI, but it does present critically important deductive and symbolic AI concepts that are necessary for a deeper understanding of ML and GenAI. BMIN 5200 is the first in a 3-part course series on artificial intelligence, followed by BMIN 5210 (AI 2: Machine Learning) and BMIN 5220 (AI 3: Natural Language Processing).

Course objectives

Upon completion of this course, students will be able to:

- Describe the unique challenges of representing and applying human knowledge in an *in silico* environment
- Explain the conceptual differences between deduction and induction, and the role of both in AI

- Discuss the past and current challenges to developing and implementing AI in biomedical domains
- Demonstrate knowledge of how common AI approaches can be developed to address real-world biomedical problems
- **Implement** core AI methods in Python — inference engines, search algorithms, expert systems, and probabilistic models — rather than only describing them

Prerequisites

It is expected that all students will be somewhat familiar with basic biomedical concepts and terminology, as well as statistics and computer programming. It is suggested (but not strictly required) that students have taken Introduction to Biomedical Informatics (BMIN 5010), Data Science for Biomedical Informatics (BMIN 5030), and a programming course (any language is fine). All programming in this course will be done in Python, and a brief introduction to the language will be provided.

No previous exposure to artificial intelligence is assumed.

Class meetings

Thursdays, 3:30–6:30pm, in person.

3600 Civic Center Boulevard, Room 6E 031.

First meeting **August 27, 2026**. Last meeting **December 3, 2026**.

Class will not meet on:

- **October 1, 2026** — Fall Break (October 1–4)
- **November 26, 2026** — Thanksgiving (November 26–29)

That leaves 13 meetings, one per lecture topic.

Recording and virtual attendance

Class sessions are recorded over Zoom and posted to Canvas, so you can review a lecture later or catch up on one you missed. The recordings are for enrolled students only — do not redistribute them.

There is no option to attend virtually. The recordings exist for review, not as a substitute for being in the room: this course is built around discussion, journal club, and in-class work, none of which function if you are watching a video afterward. Attendance is in person.

Format of a typical meeting

0–30 min	Homework review
~60 min	Weekly lecture
~25 min	In-class coding exercise
10 min	Break
~25 min	Journal club

The in-class coding exercise is new this year, and it replaces the second journal club paper. Each week’s exercise is a Jupyter notebook that runs in Google Colab — you click a link, and it works. You will build a propositional inference engine, a forward and backward chaining rule system, a search algorithm suite, an expert system in CLIPS, a Bayesian network, and a clinical agent, among others. These exercises earn credit for engagement and completion rather than for correct answers, and they are where the course’s ideas stop being abstract. Several of them are direct on-ramps to the homework.

You must bring a laptop to every class. You do not need to install anything: the exercises run in a browser through Colab. If you would rather work locally, a `requirements.txt` is provided in the course repository.

Course units

This is a 1.0cu course.

Where course materials live

Course website: the schedule, links to every lecture deck, the in-class notebooks, the journal club papers, and the assignment specifications. This is the first place to look.

Penn Box: lecture slides, journal club PDFs, and assignment files. Box links from the course website require Penn SSO — sign in with your PennKey.

Canvas: assignment submission, grades, and the Zoom recordings of each class. Nothing else in the course depends on it — lecture materials, notebooks, and papers are all reached from the course website.

GitHub: the in-class exercise notebooks, in a public repository, each with an “Open in Colab” link.

Office hours

TBD — a poll will go out in the first week, and the time that works for the greatest number of students will be selected. One-on-one meetings with the course director or TA may be scheduled on an as-needed basis.

Course director

Joseph D. Romano, PhD, MPhil, MA (please feel free to call me Joe)

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Teaching assistant

TBD — will be announced before the first class meeting.

Student expectations

Students are expected to come to class prepared for the day's content, and to ask questions and discuss that content following the lectures. Reviewing lecture slides before coming to class is strongly recommended.

Students are expected to read all assigned materials, participate during class sessions, and complete the required assignments. This course requires the use of a laptop computer.

Student evaluation

Assignments (four, weighted equally)	40%
Final project	40%
Class participation	20%

Assignments — 40% of final grade

Four graded submissions, weighted equally. Assignments are submitted via Canvas.

Late work receives a zero. Assignment answers are reviewed in class once the deadline has passed, which makes it impossible to accept a submission after the fact — the answers are public at that point. There is no partial-credit sliding scale and no grace period. If something comes up that will stop you meeting a deadline, email Joe as soon as you know, rather than after the deadline has gone by.

Assignment	Topic	Assigned	Due
Homework 1	Logic	Sep 3	Sep 17, 11:59pm
Homework 2	Semantic networks & search	Sep 24	Oct 8, 11:59pm
Homework 3	Expert system in CLIPS	Oct 15	Nov 5, 11:59pm
Homework 4	Bayesian networks	Oct 22	Nov 19, 11:59pm

Final project — 40% of final grade

Either an AI-based research project (for doctoral students) or a knowledge-based system implementation (for master's, professional, and other students), addressing a biomedical problem of your choice. Full details below, and discussed in class. The 40% breaks down as:

Component	Weight	Due
Proposal	5%	Oct 22, 11:59pm
Implementation and code	10%	Dec 11, 11:59pm
Written report	17%	Dec 11, 11:59pm
Recorded presentation	8%	Dec 11, 11:59pm

Class participation — 20% of final grade

Component	Weight
Journal club presentation	10%
Journal club discussion	4%
In-class exercises and preparation	6%

The journal club presentation is worth as much as one homework assignment, which is roughly what it costs to prepare. Discussion credit is for the other ten weeks — everyone reads the paper, not only the pair presenting it. In-class exercise credit is for engagement and completion, not for correct answers.

Attendance is expected rather than credited. There are no points for showing up, so it is not listed above. Each unexcused absence deducts 5% from your final number grade for the course, separately from the categories above.

There is no form to fill in and no list of approved reasons. Email Joe as soon as something arises — illness, a conflict, a family situation, anything — and it will be treated as excused. What turns an absence into an unexcused one is not the reason behind it but the silence: absences become unexcused when you do not get in touch. The same goes for questions about anything else in this syllabus. Ask early.

Journal club

Each week from Week 3 onward, we dedicate roughly 25 minutes to discussing one journal paper on that week's topic. **This is half of what the course ran in previous years** — one paper per week instead of two — so that we can spend the reclaimed time writing code.

A **pair** of students is assigned to each paper. One presents background and methods, the other results and critique, and both lead the discussion afterward. With eleven papers and roughly twenty students, everyone presents exactly once.

Paper presentations should include an overview of the paper (background, methods, results, and the authors' conclusions), followed by a set of discussion questions posed to the rest of the class.

Presenters then lead the class in discussion and critique.

All students are expected to have read the paper and to participate actively in the discussion. All papers are available in PDF format through the course website. The subject matter of each paper is briefly discussed in class two weeks before the presentation date, to allow students time to prepare.

LLM-driven critique

As part of each presentation, presenters must run the paper past an LLM chatbot of their choice, ask it to critique the work, and report back on: what it flagged, which of those criticisms are actually correct, and which are confidently wrong. The third question is the important one.

Final project

The final project ties together multiple topics covered in this course, with a specific focus on one or more sources or types of biomedical data. There are two types of final project, selected based on your degree program or registration status.

Research project (doctoral students; others may opt in)

Students use the concepts taught over the semester to complete an original research project addressing an area of interest or need in medicine. Examples include designing and evaluating a new clinical decision support algorithm, developing an ontology or knowledge base to support a specific domain along with use cases, or applying Bayesian networks to accomplish a novel task on a biomedical dataset. All projects should be applied to either a real or synthetic dataset, and should be cleared with Joe before work begins.

Note that **deductive inference must be used** in this project. A project that exclusively or primarily focuses on machine learning is not acceptable.

Implementation project (master's and professional students)

Students propose and implement their own software tool to address a target biomedical task — for example, an expert system that makes deductions toward a clinical task, such as a decision support system, an alert or reminder system, or one that proposes a diagnosis.

Students conduct a literature search targeting a specific biomedical task or problem, identify relevant data, databases, experts, and resources for developing the tool, and complete an initial implementation. The tool does not have to be production-ready, but should demonstrate the intended functionality. This tool should be implemented in CLIPS (e.g., via clipspy).

Proposal

A one-page proposal is due **October 22, 2026, 11:59pm**, and is worth 5% of the final grade. It states the biomedical problem, where deductive inference enters, the data or resources you plan to use, and which of the two project types you are doing. This is how your topic gets cleared before you start building. It falls the week after CLIPS is covered in class, so implementation students propose a tool having already worked with it.

Final report

Both project types involve a report with the same sections. The report should be an 8–10 page description of the project, including:

- **Background** — synthesize relevant literature and the need for AI-based approaches to the problem. Include descriptions of any existing or previous deductive AI solutions in the area.
- **Materials and Methods** — data, databases, and other resources used by and for the creation of the system.
- **Design** — the high-level design of the proposed tool and how it integrates multiple topics covered in this course.
- **Demonstration** — application of the tool to your task of interest and the results. If the tool is not completely implemented, or does not function as intended, discuss the current stage of implementation and your troubleshooting attempts.
- **Discussion and Conclusions** — challenges you overcame, advantages and disadvantages of the tool, and how it could be improved or applied more broadly.
- **References** — relevant citations. No specific format is required, but consistency and completeness both matter. Zotero or similar citation management software is recommended.

The report must be typeset in L^AT_EX using the [AAAI template](#).

Code submission. Everyone additionally submits the code accompanying their project, either (preferred) in a GitHub repository linked from the text of the paper, or as a documented, well-structured directory alongside the report, compressed into a ZIP file.

Oral presentation

A 10-minute presentation structured around the format of the written report. Slides will be helpful. The presentation is recorded as a video and uploaded to the Box folder linked from the course website.

Your slides should include a brief introduction to the biomedical problem, how you approached solving it using a deductive reasoning approach, and images (or a live demonstration) showing your program’s input and output. Research projects should also briefly describe the evaluation approach and key takeaways.

Please stick to the 10-minute limit. Since you can edit and re-record, there is no reason to run over, and points may be deducted for videos that run long.

Both the report and the presentation video are due December 11, 2026.

Course materials

No textbooks are strictly necessary, but the following are recommended:

- Cawsey A. *The Essence of Artificial Intelligence*. Pearson, 2nd ed., 2010.
- Russell S and Norvig P. *Artificial Intelligence: A Modern Approach*. Pearson, 4th ed., 2020.

Both may be found affordably online.

Academic honesty

All work submitted for credit is expected to be your own work. In the preparation of all papers and other written work, you should always take great care to distinguish your own ideas and knowledge from information derived from other sources. The term “sources” includes not only published primary and secondary material, but also information and opinions gained directly from other people. The responsibility for learning the proper forms of citation lies with you. You must acknowledge any collaboration and its extent in all submitted work. You are expected to follow Penn’s standards of academic integrity as found in the [Code of Academic Integrity](#).

Violations

All violations of academic honesty and integrity are taken very seriously and will result in disciplinary action. Depending on the seriousness of the violation, action will be taken at the discretion of the course director, and may include reduction of grades, additional assignments, a failing grade for the course, and/or reporting the incident to the University for further evaluation.

Statement on the use of artificial intelligence in assignments

Although this is a course on AI, all work completed as part of the assignments, journal club, and final project must be completed on your own volition, without the assistance of generative AI tools (or similar), unless otherwise instructed by the course director. Any and all violations of this policy will be treated similarly to other forms of plagiarism.

Two exceptions:

1. The **LLM-driven critique** portion of each journal club, in which you are expected to use the LLM chatbot of your choice to assist in critiquing the article.
2. The **in-class coding exercises**, which earn credit for engagement rather than for correct answers. Use whatever tools help you learn there — though you will get more out of them by typing the code yourself.

The homework assignments and the final project — proposal, code, report, and presentation — are covered by the policy without exception.

Please direct any and all questions to Joe — if you have any doubts, it is better to ask in advance.

Students with disabilities

The University of Pennsylvania provides reasonable accommodations to students with disabilities who have self-identified and been approved by the office of Student Disabilities Services (SDS). Please make an appointment to meet with me as soon as possible in order to discuss your accommodations and your needs. If you have not yet contacted SDS and would like to request accommodations or have questions, you can make an appointment by calling (215) 573-9235. The

SDS office is located in the Weingarten Learning Resources Center at Stouffer Commons, 3702 Spruce Street, Suite 300. All SDS services are free and confidential. Please visit the [SDS website](#) for more information.

Course materials: reuse and redistribution

Course materials fall into two groups, and they are not treated the same way.

Public materials — reusable. Everything in the public course repository on GitHub — the in-class notebooks, the course website, and the structure and organization of the course itself — is released under a [Creative Commons Attribution-NonCommercial-ShareAlike 4.0](#) licence. You may share and adapt this material, including after the course ends, provided you give attribution, do not use it commercially, and license anything you build from it on the same terms. Instructors elsewhere are welcome to it.

Anything behind a Penn login — not redistributable. Material hosted on Canvas or Penn Box, including the lecture slides, the journal club papers, the assignment files, and the class recordings, is available to enrolled students only and must not be redistributed. This is not a formality: much of that material is copyrighted by publishers and others, and the course director holds neither the copyright nor a licence to pass it on. Posting a journal club PDF to a public repository or a file-sharing site is a copyright violation regardless of the course's own licensing.

If you are unsure which group something belongs to, the rule of thumb is where you got it: if PennKey was required, treat it as not redistributable.

University and program policies

This syllabus covers the policies specific to BMIN 5200. Policies that are not addressed here are governed by the [IBI Master's and Certificate Programs Student and Faculty Handbook](#) (2026–2027 edition), which is the reference for anything this document does not say. Among other things, it covers the Code of Academic Integrity and Code of Student Conduct, the University's Equal Opportunity and Nondiscrimination Statement, the Sexual Misconduct Policy and associated resource offices and complaint procedures, student and academic grievance procedures, registration, leave of absence, audit, academic standing and probation, the program's standardized grading scale, course evaluations, the policy on secular and religious holidays, suspension of normal operations and University closures, guidance on academic issues for students with disabilities, and University resources including Counseling and Psychological Services (CAPS) and the Weingarten Learning Resources Center.

The handbook is available at:

<https://www.med.upenn.edu/BMI-training-programs/student-and-faculty-handbook.pdf>

Note that the handbook governs the Biomedical Informatics master's and certificate programs (MCI, MSBMI, and the BMI Certificate). Students enrolled through a doctoral program or another school should follow their own graduate group's policies where the two differ; the course-specific policies in this syllabus apply to everyone.

Schedule at a glance

Journal club papers are deliberately one week behind the lecture they cover: the paper discussed in a given week is on the previous week's topic, so that presenters have already been taught the material they are presenting on. Weeks 1 and 2 have no paper.

#	Date	Topic	Journal club	Due
1	Aug 27	Course intro; history of AI	—	
2	Sep 3	Knowledge representation & logic	—	
3	Sep 10	Semantic networks, frames, ontologies	Babalou et al.	
4	Sep 17	Heuristic, local & population-based search	Wolpert & Macready	HW 1
5	Sep 24	Biologically-inspired search	Nagarajan & Babu	
—	Oct 1	<i>No class — Fall Break</i>		
6	Oct 8	Rules & knowledge-based systems	Shortliffe et al.	HW 2
7	Oct 15	Building an expert system: CLIPS / clipspy	Michalowski et al.	
8	Oct 22	Bayesian networks; state machines	Leclerc et al.	Project proposal
9	Oct 29	Information theory & machine learning	Shen et al.	
10	Nov 5	Deep learning & large language models	Vaswani et al.	HW 3
11	Nov 12	Explainable AI	Behrad et al.	
12	Nov 19	Bias & fairness in AI	Pfohl et al.	HW 4
—	Nov 26	<i>No class — Thanksgiving</i>		
13	Dec 3	Agentic AI	Thirunavukarasu et al.	
—	Dec 11			Final report + video